Summary

In the twentieth century, one of the landmark discovery in biology was the identification of DNA, as genetic material of living organisms. Gene may be defined as a segment of DNA which is responsible for inheritance and expression of a particular character.

In 1953, James Watson and Francis Crick proposed DNA structure based on X-ray crystallographic studies provided by Maurice Wilkins and Rosalind Franklin. Nucleotides are the structural units of nucleic acids. Each nucleotide has three components, i) pentose sugar ii) nitrogenous base and iii) phosphate. DNA and RNA are polynucleotides. DNA has double stranded helical structure while RNA is a single stranded structure. DNA acts as genetic material of almost all the living organism except few viruses.

The non genetic RNAs are of three types; m-RNA, r-RNA and t-RNA. They help in protein synthesis. DNA has capacity of replication, while the three types of RNA are transcribed on DNA. Meselson and Stahl (1958) proved experimentally the semi-conservative nature of DNA replication using heavy isotope of nitrogen N^{15} in *E.coli*.

In 1958 Crick proposed that DNA determines the sequence of amino acids in a polypeptide (protein) through mRNA, and proposed the central dogma of protein synthesis which involves transcription and translation. The process of copying genetic information from one strand of DNA into RNA is termed transcription. The DNA transcribed RNA molecules serve as a template for the synthesis of polypeptides by a process termed translation. Each amino acid in a polypeptide chain is represented by a sequence of three nucleotides in the RNA known as the genetic code. RNA transfers genetic message from nucleus to the cytoplasm. DNA is always present in the nucleus and synthesis is also confined to the nucleus

Jacob and Monod proposed the classical model of *Lac* operon to explain gene expression and regulation in *E. coli*. In *lac* operon a polycistronic structural gene is regulated by a common promoter and regulator. It is an example of negative control of transcription initiation.

Human genome project, a mega project was aimed to sequence every gene in the human genome. Polymerase chain reaction is an *in vitro* method of synthesis of nucleic acids wherein, a specific DNA segment is amplified rapidly without concomitant replication of the rest of the DNA molecule. DNA fingerprinting is a technique to identify variations in individuals of a population at the DNA level. It has immense applications in the field of forensic analysis, pedigree analysis, anthropological studies, and conservation of wild life.

Evaluation



1. Hershey and Chase experiment with bacteriophage showed that
   a) Protein gets into the bacterial cells
   b) DNA is the genetic material
   c) DNA contains radioactive sulphur
   d) Viruses undergo transformation

2. DNA and RNA are similar with respect to
   a) Thymine as a nitrogen base
   b) A single-stranded helix shape
   c) Nucleotide containing sugars, nitrogen bases and phosphates
   d) The same sequence of nucleotides for the amino acid phenyl alanine

3. A mRNA molecule is produced by
   a) Replication **b) Transcription**
   c) Duplication d) Translation

4. The total number of nitrogenous bases in human genome is estimated to be about
   a) 3.5 million b) 35000
   c) 35 million **d) 3.1 billion**

5. *E. coli* cell grown on ^{15}N medium are transferred to ^{14}N medium and allowed to grow for two generations. DNA extracted from these cells is ultracentrifuged in a cesium chloride density gradient. What

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